

Related Projects — Comparative Research

The Capgemini MAATR case study, and the gaps left by every comparable project

Citation integrity note. Every claim below is traced to a source I fetched, cited inline. Vendor-reported and self-reported figures are labelled as such. Claims that could not be verified are listed in §7 and are **not used** in the analysis.

1. An important correction at the outset

The project referred to as "**Matri**" is in fact **MAATR** (Sanskrit for *mother*). Everything else in the description checks out: Capgemini, Maharashtra, ASHA workers, maternal health, a mobile app.

Two names must **not** be confused with it: - **PMMVY** (*Pradhan Mantri Matru Vandana Yojana*) — an unrelated central **cash-transfer** scheme. - **M-SAKHI** — a *different* Maharashtra ASHA app which, unlike MAATR, does have a published evaluation.

And the headline answer to the question "did it fail?" is: no. MAATR did not fail. It is, on the available record, Capgemini's most successful CSR technology project in this space, and it is currently scaling across Maharashtra. The interesting finding is not failure — it is *what MAATR still has not done*, and that is where ASHAMitra's opening lies.

2. Case Study — MAATR (Capgemini · Maharashtra)

2.1 Origin

MAATR began as an entry in **Capgemini's global Tech4Positive Futures Challenge (2020)**, which it **won**, beating **162 ideas from 12 countries** and receiving a **€100,000** grant. The team (Mayank Tiwari, Pulkit Dublsh, Saswat Mohapatra) spent **12+ months** researching India's RMNCH sector before building. Sources: <https://www.capgemini.com/in-en/news/inside-stories/maatr-app/> · <https://www.capgemini.com/in-en/news/inside-stories/tracking-maternal-health/>

2.2 Trajectory and scale

Stage	What happened
Pilot (Karnataka, <i>not</i> Maharashtra)	PHCs in Bengaluru and rural Karnataka, with NGO Karuna Trust ; ~90 health workers
Maharashtra agreement	Agreement with the Government of Maharashtra ; implementing partners Agriculture Development Trust (ADT) , Baramati and its arm IICARE Foundation , "under the guidance of NHM Maharashtra "
Launch (Jan 2025)	350 ASHA workers in Baramati Taluka
Scale-up	→ ~30,000 users ; stated target 1.25 lakh users by March 2026 , covering ASHAs, ANMs, ASHA Facilitators and Health Officers ; 50,000+ daily transactions
Today	Google Play: MAATR , developer IICARE Foundation , 100,000+ downloads , v5.0.3 (Aug 2025) — actively maintained

Sources: <https://play.google.com/store/apps/details?id=com.maatr> · <https://iicare.org/> · scale figures via a LinkedIn post by **Shrikant Shete, Anka Technology Solutions** (an apparent development partner) — **single-source and self-reported; treat as strong but not independently audited.**

Hard documentary confirmation comes from Capgemini's own CSR board documents, which name MAATR verbatim:

"Scaleup of Project MAATR to provide solutions to ASHA workers which will enable them to identify and mitigate risky pregnancies" — Capgemini **CSR Plan FY 2023-24**, <https://prod.ucwe.capgemini.com/in-en/wp-content/uploads/sites/18/2023/12/CSR-Plan-FY-2023-24.pdf> (the same line item also appears in the **FY 2022-23** plan, <https://www.capgemini.com/in-en/wp-content/uploads/sites/6/2022/08/CSR-Plan-FY-2022-23.pdf> — i.e. a **multi-year** funded commitment, not a one-off).

*Budget caveat: the rupee figures sit on a combined row covering both "Project FARM" and "Project MAATR." No MAATR-specific budget should be quoted.***

2.3 Technology and approach

Stated stack: **predictive analytics, AI/ML risk-case predictors, remote diagnostics, voice technology**, teleconsultation, scheduling, and an **e-learning platform** for ASHA upskilling. Delivery includes distribution of **tablets and Point-of-Care (POC) diagnostic devices** to ASHAs.

Official description (Google Play, verbatim):

"...primarily equips the on-ground ASHA workers to capture comprehensive health data of expecting mothers, identifying high risk pregnancies, facilitating medical intervention and offer guidance on Antenatal, neonatal and postnatal care. The app aims to digitize the otherwise manual effort of ASHA workers and flag high risk cases facilitating timely medical support."

2.4 Verdict: succeeded on adoption — but sits in an evidence vacuum

✔ **What succeeded.** By every *adoption* measure MAATR is a genuine success: from 90 workers to ~30,000 users, 100k+ downloads, 50k daily transactions, an active release cadence, a state-government agreement, and multi-year CSR funding. It did not fail, was not discontinued, and faced no recorded ASHA opposition.

⚠ **What it has not established.** There is **no independent or peer-reviewed evaluation of MAATR** — no RCT, no quasi-experimental study, **no published MMR, IMR or NMR impact data of any kind**. Every impact claim traces back to Capgemini, IICARE, or the vendor. MAATR also does **not appear anywhere in Capgemini Technology Services India Ltd's Annual Report FY 2024-25** (192 pages; zero hits for "maatr", ASHA, maternal, or IICARE — <https://www.capgemini.com/in-en/wp-content/uploads/sites/18/2025/08/Annual-Report-2024-25.pdf>). *That is not proof of discontinuation — impact assessment is only mandatory for CSR projects above ₹1 crore, and MAATR's line was below that and shared. But it does mean Capgemini publishes no audited outcome data on its own flagship health project.***

The one-sentence verdict: MAATR proved that ASHAs will adopt a maternal-health app at scale. It has not proved that doing so saves a single mother's life. Adoption ≠ impact.

2.5 The steps MAATR followed to succeed — and which ASHAMitra should copy

These are the replicable strategies, in rough order of how load-bearing they appear:

- 1. **A local implementing partner with real institutional and political weight.** Not a generic NGO — **ADT Baramati** (deep rural roots) plus IICARE. This is almost certainly what converted a Karnataka pilot into a state-scale Maharashtra rollout. (*ASHAMitra's equivalent: AIIMS Kalyani + the West Bengal NHM.*)
- 2. **Government integration from the top, not a parallel silo.** Framed as running *under NHM Maharashtra's guidance*, with a state-government agreement — so it becomes **the** official workflow rather than an extra app. **This is the difference between life and death for an ASHA app.**
- 3. **Start narrow, prove it, then scale.** 90 workers → 350 in one taluka → 30,000. A disciplined ladder, never a big-bang launch.
- 4. **Onboard the entire hierarchy, not just ASHAs.** The rollout deliberately includes **ANMs, ASHA Facilitators and Health Officers**. *This is the single most important anti-failure move.* If only ASHAs use the app while supervisors still demand paper, you get double entry and abandonment. Onboarding the supervisory chain is what actually kills the paper register.
- 5. **Hardware provided, not assumed.** Tablets and POC devices distributed free — removing the "use your own phone, your own data" resentment that has destroyed other ASHA apps.
- 6. **Explicit burden-reduction framing.** The stated goal is to "*digitize the otherwise manual effort of ASHA workers*" — **replacing** registers, not adding to them.
- 7. **Clinical value returned to the worker.** High-risk flagging, ANC/PNC guidance, plus e-learning — the worker *gets something back* rather than merely feeding a government database.
- 8. **Long-horizon funding.** Carried as an ongoing line item across at least two CSR years — no cliff after year one.

2.6 The gaps MAATR has not filled — the white space for ASHAMitra

#	Gap in MAATR	Why it matters
1	No published clinical-outcome evidence. Zero peer-reviewed evaluation; no MMR/IMR data.	It is the same pathology as the whole field: scale ahead of evidence .
2	Black-box logic. MAATR's stated approach is " predictive analytics / AI-ML risk predictors ."	An AI/ML risk model cannot be read, approved or certified by a clinical committee , cannot be reproduced months later for an MDSR audit, and cannot cite a guideline per decision. This is precisely the property AIIMS will scrutinise.
3	No published clinical-governance mechanism. No documented committee sign-off gate, no rule-versioning/tamper seal, no auditable per-case decision trace.	Without this there is no clinical accountability chain.

#	Gap in MAATR	Why it matters
4	Hardware dependency. Tablets + POC diagnostic devices distributed.	Capital-intensive; scaling to 1.25 lakh users implies enormous recurring hardware and support cost — a sustainability risk if CSR funding tapers (and MAATR is absent from the FY24-25 annual report).
5	No evidence of referral-loop closure. Nothing in the public record shows MAATR tracks whether a referred mother actually reached the facility.	This is the gap e-Sanjeevani's own evaluators call " <i>a critical gap</i> ."
6	Wrong geography and language. Maharashtra/Marathi.	West Bengal — with rising MMR and 62.3% pregnancy anaemia — has no ASHA-facing app at all , in a language digital health has neglected.

Conclusion of the case study: MAATR is not a cautionary tale — it is a **proof of demand** and a **playbook of go-to-market strategy** (steps 1–8 above). Its unfilled space is **clinical governance and evidence**: deterministic, auditable, guideline-cited, committee-signed logic, plus a real evaluation. That is exactly what ASHAMitra is built to be.

3. Other comparable projects — and their gaps

3.1 ImTeCHO / TECHO+ (Gujarat) — the best-evidenced ASHA job aid in India

Cluster RCT, 22 PHCs, 5,754 live births (Modi et al., *PLOS Medicine* 2019, <https://journals.plos.org/plosmedicine/article?id=10.1371/journal.pmed.1002939>). - **Achieved:** MNCH coverage index **+4.9 pp** ($p=0.03$); ≥ 2 first-week home visits **+10.2 pp**. - **GAP — no mortality effect.** 233 vs 236 infant deaths; 104 vs 102 neonatal deaths. *Nothing*. - **GAP — parallel paper survived even at scale.** The TECHO+ state evaluation records that "*some ANMs maintained dual records (paper and digital) due to offline operation limitations*," with 38% reporting repeated-entry problems (<https://www.frontiersin.org/journals/public-health/articles/10.3389/fpubh.2022.856561/full>). - **GAP — supervisory adoption failed.** Only **3 of 11** PHCs used the automated incentive system; web-interface adoption by PHC staff was low.

⚠ Citation hazard. The widely-quoted "*ImTeCHO averted 11 infant deaths per 1,000 — a 16% reduction*" comes from the **per-protocol analysis of the cost-effectiveness paper** (<https://pmc.ncbi.nlm.nih.gov/articles/PMC7593859/>), **not** the trial's intention-to-treat result. **The RCT found no mortality effect.** Do not cite the 16% figure.

3.2 Khushi Baby / CHIP (Rajasthan) — scale far ahead of evidence

- **The one rigorous trial was NULL.** Cluster RCT, 96 village health camps, Udaipur: the interventions "*did not directly result in an increase in infant immunization timeliness through DTP3*" — not statistically significant (Nagar et al., *Vaccine* 2018, <https://pubmed.ncbi.nlm.nih.gov/29162321/>).
- **GAP:** Despite the null trial, CHIP scaled to state level (self-reported $\approx 70,000$ CHWs, 40–46 M beneficiaries; ₹107.5 crore MoHFW funding) — **with no published evaluation of CHIP at state scale.** A textbook case of scaling on the strength of a null result.

3.3 Kilhari + Mobile Academy (national, MoHFW/BBC Media Action) — the largest, and largely null

- **Kilhari RCT (Madhya Pradesh): primary outcome NULL** — exclusive breastfeeding RR 1.04 (95% CI 0.88–1.23, $p=0.64$). Small significant secondaries: contraception **+3.7 pp**, immunisation at 10 weeks **+2.8 pp** (Mohan et al., *BMJ Global Health* 2022, <https://pmc.ncbi.nlm.nih.gov/articles/PMC9288869/>).
- **GAP — the government data pipeline is the weak link, and it is quantified.** Of 13.6 M records, **70% were rejected at validation** (46% invalid pregnancy dates, 40% duplicate numbers). **Only 21% of pregnant women** across 13 states actually received Kilhari (Delhi 77% → Bihar 3% — lowest reach where need is highest). **37% of subscriptions started after childbirth.** **Only 31% had a birth-date update**, so $\sim 69\%$ received stage-misaligned content (<https://pmc.ncbi.nlm.nih.gov/articles/PMC9366343/>).
- **GAP — Mobile Academy reach:** only **24%** of registered ASHAs nationally initiated the course; **West Bengal initiation: 1%** (vs Rajasthan 64%) (<https://pmc.ncbi.nlm.nih.gov/articles/PMC8386225/>).
- **GAP — equity:** under-represents marginalised castes and women without education; **44% of women changed SIM cards** during the trial.

3.4 mMitra (ARMMAN, Mumbai)

- Pseudo-randomised (not truly random) trials showed improved knowledge and practices; birth-weight "trend" only.
- **GAP — the authors themselves concede it.** Their methods paper is titled "*The Elusive Path Toward Measuring Health Outcomes: Lessons Learned From a Pseudo-Randomized Controlled Trial...*" (<https://pmc.ncbi.nlm.nih.gov/articles/PMC6724498/>).
- **GAP — it is a beneficiary channel.** It does nothing for the ASHA's decision-making at the point of care.

3.5 ReMiND (CARE India, UP) — the most important comparator, because it has decision support

CommCare-based ASHA job aid with **counselling prompts, decision support and simple referral algorithms** (Prinja et al., *Cost Eff Resour Alloc* 2018, <https://pmc.ncbi.nlm.nih.gov/articles/PMC6020234/>). - **Achieved** — and note *where*: **complication identification +13.2% (pregnancy), +19.5% (postpartum)**; care-seeking **+25.7%**; IFA **+12.7%**. Cost-effective at ₹12,993/DALY averted. - **This is the single most useful data point in the entire literature for ASHAMitra**: ReMiND's biggest gains were in **complication identification — a decision-support function — not in service utilisation (institutional delivery: no change)**. - **GAP — mortality was modelled, not measured**. - **GAP — it assumes the referral completes**. The authors explicitly flag that the model "assumes perfect supply-side capacity to meet increased demand." The loop is never closed.

3.6 ICT-CCS (CARE India, Bihar) — cluster RCT

- **Achieved**: first-week postnatal visits 72% vs 60%; skin-to-skin **+13 pp**; immediate breastfeeding **+12 pp**; complementary feeding **+21 pp**.
- **GAP: no effect** on facility delivery, immunisation, exclusive breastfeeding — **or supervision** (<https://pmc.ncbi.nlm.nih.gov/articles/PMC6875677/>).

3.7 ANMOL (national, MoHFW/UNICEF) — the flagship with no evidence

- **GAP — no impact evaluation exists**. The national ANM tool has essentially **no published effectiveness evidence**.
- **GAP — experienced as surveillance and duplication**. The only peer-reviewed qualitative study (all 24 ANMs, Salon block, Raebareli) found poor connectivity, technical glitches and "**duplication in data entry**", with dominant themes of "**increased work**" and "**monitoring of the activities**" (*Indian J Community Med* 2026;51(2):358–365, https://journals.lww.com/ijcm/fulltext/2026/03000/perception_and_experiences_of_auxiliary_nurse.21.aspx).
- **GAP — no clinical decision support**. It is a registration/reporting system.

3.8 Poshan Tracker / ICDS-CAS (Anganwadi, MoWCD)

- **ICDS-CAS (evaluated)**: improved home visits **+8 pp** and counselling recall — but **no effect on infant/young-child feeding practices** and, tellingly, **no improvement in actual receipt of take-home rations** despite better counselling (Avula et al., *BMJ Global Health* 2022, <https://pmc.ncbi.nlm.nih.gov/articles/PMC9296874/>). *The classic open-loop signature: the message improves, the service does not arrive*.
- **Poshan Tracker (current)**: **no peer-reviewed impact evaluation found**, but well-documented harm — workers reduced to "**data clerks**"; **Aadhaar-linked facial recognition made mandatory for take-home rations (from July 2025), excluding eligible pregnant women when it fails to match** (<https://www.boomlive.in/decode/ai-facial-recognition-is-denying-food-to-pregnant-women-across-india-30878>); ~10,000 workers protesting in Karnataka.
- **GAP**: a digital system that **actively excludes** the beneficiary it was built to serve.

3.9 e-Sanjeevani / ABDM (national backbone)

- **GAP — stated by the evaluators themselves**. The e-Sanjeevani adoption analysis concedes it "*does not assess how eSanjeevani integrates with downstream elements of primary care — such as access to medicines, diagnostics or in-person referrals,*" and that "*whether patients are able to complete treatment plans initiated remotely remains a critical gap*" (<https://pmc.ncbi.nlm.nih.gov/articles/PMC12558045/>). **This is the referral-loop gap, named in the national programme's own evaluation**.
- **GAP — HMIS data is not fit for clinical use**. Validation in Haryana (209 sub-centres, 4,807 women) found **only 26.6% of ANC visit counts and 20.1% of postnatal visit counts recorded correctly** at individual level; IFA supplementation over-reported by 41% (*PLOS ONE* 2016, <https://journals.plos.org/plosone/article?id=10.1371/journal.pone.0148449>).

3.10 E-IMNCI (Jharkhand) — the closest thing to ASHAMitra's logic, and it worked

A digitised, deterministic IMNCI algorithm (adapted from leDA, West Africa) with Jharkhand NHM; 70 ANMs, 55 facilities (<https://pmc.ncbi.nlm.nih.gov/articles/PMC10603548/>). - **Achieved — dramatically**: respiratory-rate assessment **18% → 100%**; umbilical sepsis check **9% → 100%**; consultation time **24 → 12 min**; 91% of ANMs found digital easier than paper. - **GAP — and this is the opening**: it is a **small pre-post pilot with no control**, and crucially it serves **facility-based ANMs/CHOs — not ASHAs at the household**. No clinical or mortality outcomes.

This is the empty cell in the literature: there is no published, evaluated, deterministic IMNCI/HBNC decision-support system running in an ASHA's hand, at the household, at scale.

4. The graveyard — what actually kills an ASHA app

Two recent, documented deaths. Both are instructive, and neither was a technology failure.

- **Hajeri (Maharashtra)** — a photo-backed attendance app. **Discontinued after ASHA protests**. ASHAs reported app proliferation, failed uploads, and unpaid data labour (<https://behanbox.com/2025/04/06/why-a-photo-backed-attendance-app-is-distressing-maharashtras->

asha-workers/).

- **MDM360 Shield (Haryana)** — a location-tracking app. Perceived as **surveillance**, created "*anxiety and distrust*," and was **recalled after collective action by ASHAs** (<https://www.ideasforindia.in/topics/productivity-innovation/is-digitalisation-a-double-edged-sword-for-workers-in-indias-public-healthcare-system>).

The rule this establishes: an app that **extracts** from the ASHA (attendance, location, reporting) without **giving her something back** will be resisted and can be withdrawn. An app that becomes an *additional* register instead of a *replacement* one dies. **MAATR survived precisely because it framed itself as burden-reduction and returned clinical value to the worker.** ASHAMitra must do the same, only more so.

This is not hypothetical in West Bengal: the state's ASHAs began **indefinite cease-work on 23 December 2025**, and among their explicit demands was "**exclusion from work unrelated to maternal and child health support**" (<https://theprint.in/india/unhappy-with-pay-hike-asha-workers-protest-outside-bengal-health-department-hq/2847628/>).

5. Comparison matrix — ASHAMitra against the field

Capability	ANMOL	Poshan Tracker	Kilkari	Khushi Baby	ImTeCHO	ReMiND	E-IMNCI	MAATR	ASHAMitra
In the ASHA's hand (not ANM/beneficiary)	✗ (ANM)	✗ (AWW)	✗ (mother)	✓	✓	✓	✗ (ANM)	✓	✓
Clinical decision support	✗	✗	✗	✗	partial	✓	✓	✓	✓
Deterministic (not black-box AI)	n/a	n/a	n/a	n/a	✓	✓	✓	✗ (AI/ML)	✓
Every rule cites a guideline	✗	✗	✗	✗	?	?	✓	✗	✓
Committee sign-off gate (refuses unsigned rules)	✗	✗	✗	✗	✗	✗	✗	✗	✓
Auditable per-case trace (protocol hash)	✗	✗	✗	✗	✗	✗	✗	✗	✓
Works fully offline	partial	✗	n/a	✓	partial	✓	?	?	✓
Replaces paperwork (auto HMIS/MCP)	✗ (dual entry)	✗ (dual entry)	n/a	✗	✗ (dual entry)	✗	✗	✓ (claimed)	✓
Referral loop closed (outcome + refusal reason)	✗	✗	✗	✗	✗	✗	✗	✗	✓
Bengali / West Bengal	✗	✗	✗	✗	✗	✗	✗	✗	✓
Published clinical-outcome evidence	✗	✗	✓ (null)	✓ (null)	✓ (null on mortality)	partial (modelled)	✗	✗	✗ — the pilot's purpose

Legend: ✓ present · ✗ absent · ? not documented. The final row is deliberately honest: ASHAMitra has no outcome evidence yet — building it is the point of the AIIMS pilot.

6. Conclusion — the defensible white space

Three things are now established by the evidence, and together they define ASHAMitra's position.

First, demand is proven. MAATR took an ASHA maternal-health app from 90 workers to ~30,000 in Maharashtra with 100k+ downloads. ASHAs *will* adopt such a tool — **if** it reduces their burden rather than adding to it. The apps that died (Hajeri, MDM360 Shield) died because they extracted without giving back.

Second, the whole field — including MAATR — shares one failure: it scales ahead of evidence, and it cannot be clinically governed. India's two largest national systems have *no* published impact evaluation. Every rigorous RCT (ImTeCHO, Kilkari, Khushi Baby, ICT-CCS) was **null** on its clinical endpoint. MAATR — the closest comparator — has **zero** published health-outcome data and is built on **AI/ML predictive analytics**, which no clinical committee can read, approve, reproduce, or defend in a maternal-death audit.

Third, the evidence points clearly at what does work, and nobody has built it for an ASHA. The interventions with the largest clinical effects were the ones with **decision support** — ReMiND's complication identification (+13.2%/+19.5%), E-IMNCI's adherence leap (18%→100%). WHO already codified this in 2019: tracking must be implemented **combined with decision support**, and tracking alone "*risks ineffectiveness*." Yet the only India CDSS work is a small pilot for **facility-based ANMs**, and India's national tools implement precisely the half WHO warned about.

The white space is therefore exact, and no existing project occupies it: a deterministic, guideline-cited, committee-signed, auditable clinical decision-support system — running offline, in Bengali, in an ASHA's hand, at the household — that replaces her paperwork rather than adding to it, closes the referral loop, and is evaluated before it is scaled.

MAATR shows the *market* is real and hands us the go-to-market playbook (government integration, whole-hierarchy onboarding, narrow→prove→scale). The literature shows the *clinical* opportunity is real and unclaimed. West Bengal — with maternal mortality **rising** against a falling national trend, 62.3% pregnancy anaemia, 65,743 ASHAs, and **no ASHA-facing app of any kind** — is where it is most needed.

What MAATR did not do is the whole of ASHAMitra's contribution: make the clinical logic something a doctor can own, and then actually prove it works.

7. Verification notes — claims deliberately excluded

1. "MAATR trained 600 ASHAs / distributed 600 tablets and 40 diagnostic kits." Appeared in search-engine summaries of a Capgemini page but was **not present in the fetched page text. Not cited.**
2. **MAATR-specific budget figures.** The CSR line item is shared with "Project FARM." **No MAATR-only rupee figure can be quoted.**
3. **MAATR scale figures (30,000 users; 1.25 lakh target; 50,000 daily transactions).** Traceable only to a **development partner's LinkedIn post** — self-reported, not independently audited. Used with that caveat stated.
4. **The Ajit Pawar / MAATR launch event.** A headline exists (<https://www.contentgarden.in/news/Ajit-Pawar-attends-the-MAATR-App-ASHA-Software-program/1915659>) but the article body would not load. **Political-patronage link is inferred, not proven — not relied upon.**
5. **NHM Maharashtra endorsement of MAATR.** Asserted by the implementing partners; **no NHM government resolution or circular naming MAATR was located** on official portals.
6. **"ImTeCHO cut infant mortality 16%."** Per-protocol figure from the cost-effectiveness paper; **the RCT found no mortality effect. Excluded.**
7. **"CommCare/Mathematica RCT: +73% ANC, +58% iron, +36% contraception."** Untraceable to any peer-reviewed source. **Excluded.**
8. **Khushi Baby / CHIP scale figures.** Organisation-reported; no independent evaluation. Labelled as such.

8. References

A. MAATR (Capgemini · Maharashtra) — primary sources

1. **Capgemini India.** *Maatr app* (inside story). <https://www.capgemini.com/in-en/news/inside-stories/maatr-app/> (*Origin, Tech4Positive Futures Challenge win, Karnataka pilot with Karuna Trust, ~90 health workers.*)
2. **Capgemini India.** *Tracking maternal health* (inside story). <https://www.capgemini.com/in-en/news/inside-stories/tracking-maternal-health/> (*Maharashtra agreement; ADT Baramati / IICARE Foundation; NHM Maharashtra.*)
3. **Capgemini Technology Services India Ltd.** *CSR Plan FY 2023-24* (PDF). <https://prod.ucwe.capgemini.com/in-en/wp-content/uploads/sites/18/2023/12/CSR-Plan-FY-2023-24.pdf> — contains verbatim: "Scaleup of Project MAATR to provide solutions to ASHA workers which will enable them to identify and mitigate risky pregnancies."
4. **Capgemini Technology Services India Ltd.** *CSR Plan FY 2022-23* (PDF). <https://www.capgemini.com/in-en/wp-content/uploads/sites/6/2022/08/CSR-Plan-FY-2022-23.pdf> (*Establishes MAATR as a multi-year funded commitment.*)
5. **Capgemini Technology Services India Ltd.** *Annual Report FY 2024-25* (PDF, 192 pp). <https://www.capgemini.com/in-en/wp-content/uploads/sites/18/2025/08/Annual-Report-2024-25.pdf> — **MAATR does not appear** (zero hits for "maatr", ASHA, maternal, IICARE). *Not proof of discontinuation, but Capgemini publishes no audited outcome data on the project.*
6. **MAATR on Google Play.** Developer: IICARE Foundation. 100,000+ downloads; v5.0.3 (8 Aug 2025). <https://play.google.com/store/apps/details?id=com.maatr>
7. **IICARE Foundation.** <https://iicare.org/>
8. **MAATR on Indus Appstore.** <https://www.indusappstore.com/apps/essential-apps/maatr/com.maatr>
9. **Scale figures** (350 ASHAs in Baramati → ~30,000 users → 1.25 lakh target; 50,000+ daily transactions): traceable only to a LinkedIn post by **Shrikant Shete, Anka Technology Solutions Pvt. Ltd.** (apparent development partner). <https://www.linkedin.com/in/shrikantshete-anka/> —  **self-reported, single-source, not independently audited.**
10. **Ajit Pawar attends the MAATR App / ASHA Software program.** <https://www.contentgarden.in/news/Ajit-Pawar-attends-the-MAATR-App-ASHA-Software-program/1915659> —  **headline only; article body did not load. Not relied upon.**

B. Comparator projects — trials and evaluations

11. **Modi D, et al.** ImTeCHO cluster randomised trial. *PLOS Medicine*. 2019. <https://journals.plos.org/plosmedicine/article?id=10.1371%2Fjournal.pmed.1002939>
12. **Modi D, et al.** ImTeCHO cost-effectiveness. *JMIR mHealth uHealth*. 2020;8(10):e17066. <https://pmc.ncbi.nlm.nih.gov/articles/PMC7593859/> —  source of the misused "16% mortality reduction" (per-protocol); the RCT (ref 11) found no mortality effect.
13. **TECHO+ state-scale evaluation.** *Frontiers in Public Health*. 2022. <https://www.frontiersin.org/journals/public-health/articles/10.3389/fpubh.2022.856561/full> (Dual paper+digital records persisted at scale.)
14. **Nagar R, et al.** Khushi Baby cluster randomised trial, Udaipur. *Vaccine*. 2018. <https://pubmed.ncbi.nlm.nih.gov/29162321/> (Null on DTP3 timeliness.)
15. **Mohan D, et al.** Kilhari randomised controlled trial. *BMJ Global Health*. 2022. <https://pmc.ncbi.nlm.nih.gov/articles/PMC9288869/> (Null on primary outcome.)
16. **Bashingwa JJH, et al.** Kilhari reach across 13 states. *BMJ Global Health*. <https://pmc.ncbi.nlm.nih.gov/articles/PMC9366343/> (70% of records rejected; only 21% of pregnant women reached.)
17. **Mobile Academy** reach and completion. <https://pmc.ncbi.nlm.nih.gov/articles/PMC8386225/> (West Bengal initiation: 1%.)
18. **Prinja S, et al.** ReMiND, Uttar Pradesh. *Cost Effectiveness and Resource Allocation*. 2018;16:25. <https://pmc.ncbi.nlm.nih.gov/articles/PMC6020234/> (Complication identification +13.2% / +19.5%; assumes "perfect supply-side capacity.")
19. **ICT-CCS cluster randomised trial**, Bihar. <https://pmc.ncbi.nlm.nih.gov/articles/PMC6875677/>
20. **Avula R, et al.** ICDS-CAS evaluation. *BMJ Global Health*. 2022. <https://pmc.ncbi.nlm.nih.gov/articles/PMC9296874/> (Counselling improved; ration receipt did not.)
21. **mMitra** trials: <https://pmc.ncbi.nlm.nih.gov/articles/PMC7268375/> · <https://pmc.ncbi.nlm.nih.gov/articles/PMC6823296/>
22. **ARMMAN.** *The Elusive Path Toward Measuring Health Outcomes...* <https://pmc.ncbi.nlm.nih.gov/articles/PMC6724498/>
23. **E-IMNCI (Jharkhand)** — deterministic IMNCI decision support. <https://pmc.ncbi.nlm.nih.gov/articles/PMC10603548/> (Respiratory-rate assessment 18% → 100%; facility-based ANMs, not ASHAs.)
24. **ANMOL — ANM perceptions and experiences.** *Indian Journal of Community Medicine*. 2026;51(2):358–365. https://journals.lww.com/ijcm/fulltext/2026/03000/perception_and_experiences_of_auxiliary_nurse.21.aspx
25. **e-Sanjeevani adoption analysis.** *Oxford Open Digital Health*. 2025. <https://pmc.ncbi.nlm.nih.gov/articles/PMC12558045/> ("Whether patients are able to complete treatment plans initiated remotely remains a critical gap.")
26. **Sharma A, et al.** HMIS/RCH data quality, Haryana. *PLOS ONE*. 2016;11(2):e0148449. <https://journals.plos.org/plosone/article?id=10.1371%2Fjournal.pone.0148449> (Only 26.6% of ANC visit counts recorded correctly.)

C. Normative source

27. **World Health Organization.** *Guideline: Recommendations on Digital Interventions for Health System Strengthening*. Geneva: WHO; 2019. <https://www.ncbi.nlm.nih.gov/books/NBK541888/> · <https://www.ncbi.nlm.nih.gov/books/NBK541902/> (Recommendation 8: tracking combined with decision support; isolated tracking "risks ineffectiveness.")

D. The "graveyard" — withdrawn apps and worker resistance

28. **Behanbox.** Why a photo-backed attendance app is distressing Maharashtra's ASHA workers, 6 April 2025. <https://behanbox.com/2025/04/06/why-a-photo-backed-attendance-app-is-distressing-maharashtras-asha-workers/> (The Hajeri app — discontinued after ASHA protests.)
29. **Sreerupa, Makkad S, Rajeev A.** Is digitalisation a double-edged sword for workers in India's public healthcare system? *Ideas for India*, 2024. <https://www.ideasforindia.in/topics/productivity-innovation/is-digitalisation-a-double-edged-sword-for-workers-in-indias-public-healthcare-system> (MDM360 Shield — recalled after collective action.)
30. **Chakraborty R.** India's digital health push is overworking its front-line women. *New Lines Magazine*, 31 March 2026. <https://newlinesmag.com/reportage/indias-digital-health-push-is-overworking-its-front-line-women/>
31. **Data & Society.** From care labor to data labor. <https://datasociety.net/points/from-care-labor-to-data-labor-indias-door-to-door-health-activists/>
32. **BOOM.** AI facial recognition is denying food to pregnant women across India. <https://www.boomlive.in/decode/ai-facial-recognition-is-denying-food-to-pregnant-women-across-india-30878> (Poshan Tracker exclusion.)
33. **ThePrint.** Unhappy with pay hike, ASHA workers protest outside Bengal health department HQ, 6 February 2026. <https://theprint.in/india/unhappy-with-pay-hike-asha-workers-protest-outside-bengal-health-department-hq/2847628/> (WB ASHA demand: "exclusion from work unrelated to maternal and child health support.")

E. Related project not confused with MAATR

34. **M-SAKHI** — a different Maharashtra ASHA application which, unlike MAATR, has a published evaluation. Noted here only to prevent conflation.

35. **PMMVY** (*Pradhan Mantri Matru Vandana Yojana*) — an unrelated central **cash-transfer** scheme. Noted here only to prevent conflation with "Matri."